# Replication Summary of Andreadis et al. (2025)

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**Abstract** Andreadis et al. (2025) argue that labor market tightness, STEM education, and local innovation explain where AI-related employment grows across U.S. counties. I was able to replicate their key results by reconstructing Tables 1 and 2 and estimating five models with demographic, innovation, and industry variables. In my replication, I reweighted the regressions using the logarithm of population to reduce the influence of large counties and reassessed coefficient significance under that design. AI continues to shape the geography of U.S. job growth, but its adoption remains uneven, and future work is needed to understand which regional factors amplify or stifle its spread.

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## Key Variables and Replication Design

The authors argue that the growth of AI-related jobs is driven by "innovation and skills, including patents, degrees per capita, and the share of STEM occupations and degrees" [Andreadis et al. (2025), p. 16]. They also highlight that "tightness in the labor market is a strong predictor of AI exposure" [Andreadis et al. (2025), p. 16]. These findings frame their emphasis on the following variables:

- Labor market tightness
- STEM degrees' share
- Patents per employee
- Bachelor's share
- Manufacturing intensity
- ICT sector intensity
- Turnover rate

Other variables, such as AI patents' share and degrees per capita, are tested but are not ultimately emphasized in the authors' interpretations.

In this replication, I re-estimate all five models from the original paper using **log(population)** weighting. This approach reduces the undue influence of high-population counties and provides a robustness check on their reported findings.

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## Figure 1: Key Coefficient Estimates for AI Share (Table 1, Log-Population Weighted)

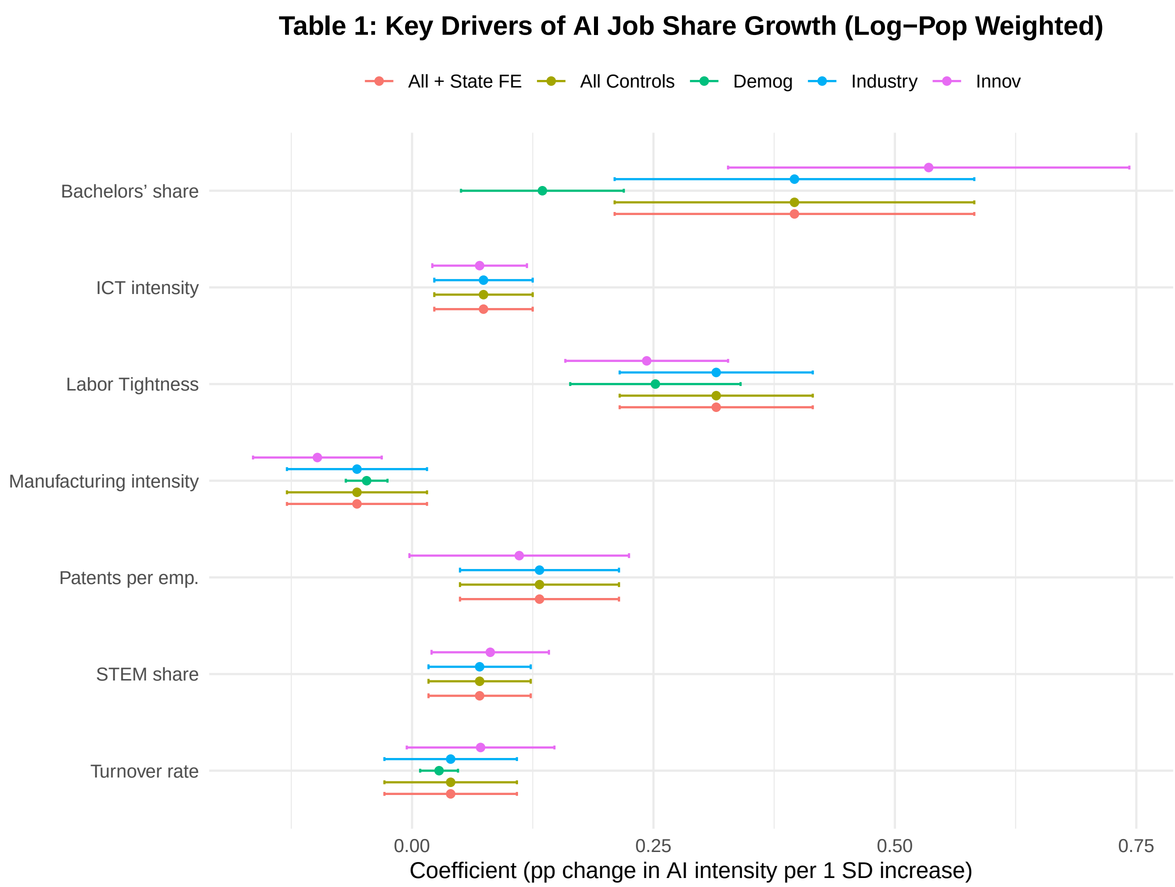


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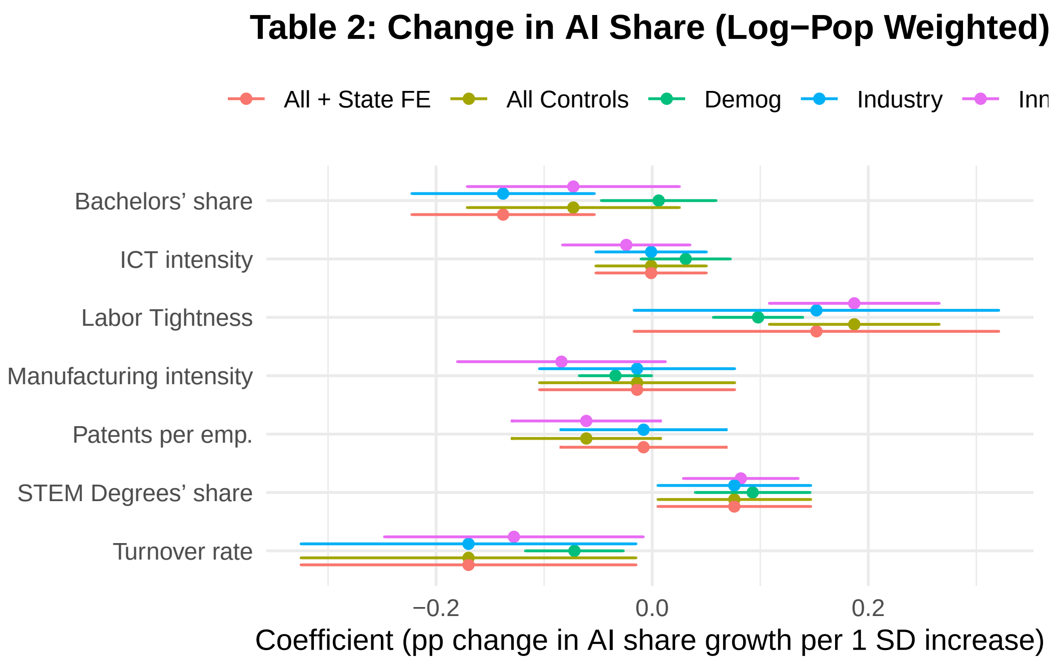
**Figure 2: Key Coefficient Estimates for Change in AI Share (Table 2, Log-Population Weighted)**

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### Interpretation of Key Results

In both Table 1 and Table 2 models, **labor market tightness** consistently predicts higher AI employment and growth. This confirms the original claim that "tightness in the labor market is a strong predictor of AI exposure" [Andreadis et al. (2025), p. 16].

**STEM education** and **innovation**—measured by patents per employee—also maintain positive associations across most models, supporting the authors' view that "innovation and skills" drive AI diffusion [Andreadis et al. (2025), p. 16].

In Table 1, the significance of **bachelor's share** and **ICT intensity** becomes stronger when reweighted by log(population), suggesting these factors may play an even more important role in less populous counties than previously emphasized.

For Table 2, the **bachelor's share** turns negative in some models, potentially challenging the assumption that highly educated areas always experience faster AI growth. While **manufacturing intensity** and **turnover rate** remain negatively associated with AI job growth, the effect sizes shift, especially in the "All + State FE" model.

These changes demonstrate how model design—especially weighting—affects variable interpretation. In particular, log(population) weighting reveals that several predictors emphasized by the authors lose statistical significance or reverse in sign when adjusting for population scale.

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## Summary

This replication confirms that AI employment is concentrated in places with high STEM education, tight labor markets, and innovation activity. Reweighting by log(population) reveals how county size skews significance and shows that some originally strong predictors weaken when accounting for population heterogeneity. While the original claims mostly hold, the shift in strength and sign for some variables—like manufacturing and turnover—suggests that future analyses should pay closer attention to weighting and scaling assumptions in local economic research.

Andreadis, Lefteris, Eleni Kalotychou, Manolis Chatzikonstantinou, Christodoulos Louca, and Christos A. Makridis. 2025. "Local Heterogeneity in Artificial Intelligence Jobs over Time and Space." *American Economic Review: Papers & Proceedings* 115 (5): xx–. https://mhatzikonstantinou.github.io/pdfs/AERp_p_paper.pdf.